

Solution Architecture

Project Information

Date	15 March 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram

The following diagram provides a high-level visual representation of the OptiCrop system architecture. The solution follows a layered structure consisting of the presentation layer, Flask backend, machine learning prediction engine, database layer, and deployment environment.

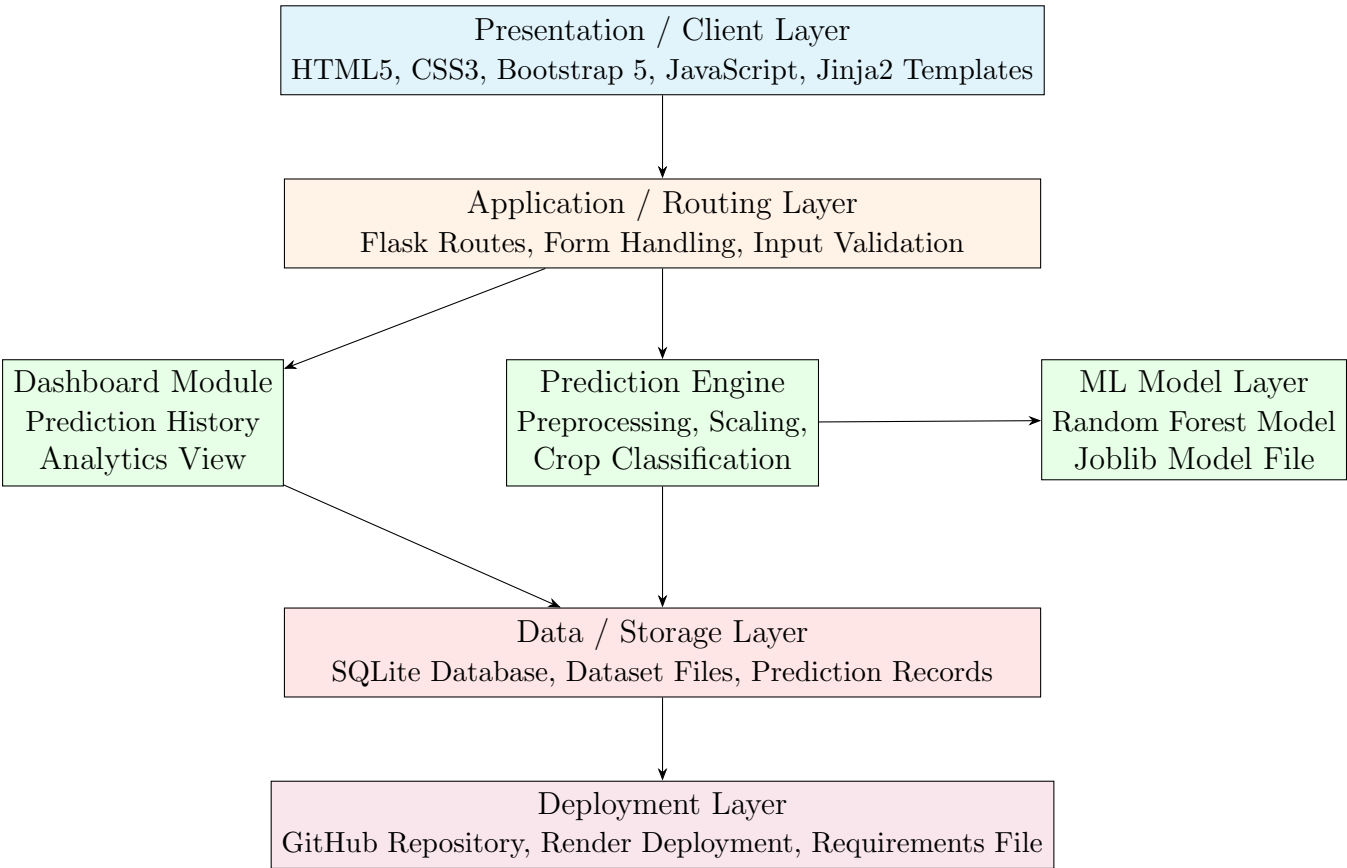


Figure 1: Solution Architecture Diagram showing the OptiCrop client interface, Flask application layer, machine learning prediction engine, SQLite storage, and deployment environment.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the web-based user interface where users enter soil nutrients and environmental values, submit prediction requests, and view crop recommendation results.	HTML5, CSS3, Bootstrap 5, JavaScript, Jinja2 Templates
Application Layer	Handles user requests, validates form inputs, routes prediction requests, manages page rendering, and connects the frontend with backend modules.	Python, Flask
Prediction Engine	Performs preprocessing, feature scaling, model loading, crop prediction, and confidence score generation for the submitted agricultural parameters.	Python, Pandas, NumPy, Scikit-Learn, Joblib
Machine Learning Model	Uses the trained model to classify the most suitable crop based on Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, and rainfall values.	Random Forest Classifier, Scikit-Learn
Database Layer	Stores prediction records, soil input values, recommended crops, confidence scores, and history information for dashboard review.	SQLite
Deployment Layer	Supports source control, dependency management, and cloud deployment readiness for demonstration and hosting.	Git, GitHub, Render, requirements.txt